Ex No: 9

Develop a Simple Calculator Using XMLRPC

Aim:

To design, implement and verify a remote simple calculator using XML-RPC in python based on client-server architecture.

Procedure:

1. Create a python server program named calcisorver.py
2. Import simpleXMLRP server from xmlrpc.server
3. Define functions of addition, subtraction, multiplication and division.
4. Check for division by zero in the division function.
5. Create the XML-RPC server on localhost using port 8000
6. Register all the functions with the server
7. Create a client program named calciclient.py
8. Connect the client to http://localhost:8000/
9. Enter two numbers and call the remote calculator functions.
10. Display the results returned by the server.

Code:

Server - calcisorver.py

```
from xmlrpc.server import simpleXMLRP
                          Server
 def add(a,b):
        return a+b
def subtract(a,b):
        return a-b
def multiply(a,b):
        return a*b
def divide(a,b):
    if b==0:
        return "Division by zero not allowed"
    return a/b
```

O/P:

py calcServer.py

XML-RPC calculator server running on port 8000.

py calcclient.py

Enter First Number: 20

Enter Second Number: 10

Addition = 30.0

Subtraction = 10.0

Multiplication = 200.0

Division = 2.0

py calcclient.py

Enter First Number: 10

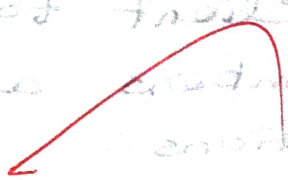
Enter Second Number: 0

Addition = 10.0

Subtraction = 10.0

Multiplication = 0.0

Division = Division by zero not allowed



```
server = SimpleXMLRPCServer(("localhost", 8000))
print("XML-RPC calculator server Running
on port 8000...")
server.register_function(add, "add")
server.register_function(subtract, "subtract")
server.register_function(multiply, "multiply")
server.register_function(divide, "divide")
server.serve_forever()
client - calaiclient.py
import xmlrpc.client
proxy = xmlrpc.client.ServerProxy("http://local
host:8000/")
a = float(input("Enter First Number:"))
b = float(input("Enter second Number:"))
print("\n Addition =", proxy.add(a,b))
print("subtraction =", proxy.subtract(a,b))
print("multiplication=", proxy.multiply(a,b))
print("Division=", proxy.divide(a,b))
```

Result:

Thus, the XML-RPC based simple calculator was successfully implemented and executed for addition, subtraction, multiplication and division